Machine Learning for Physics: Simulating Particle Collisions

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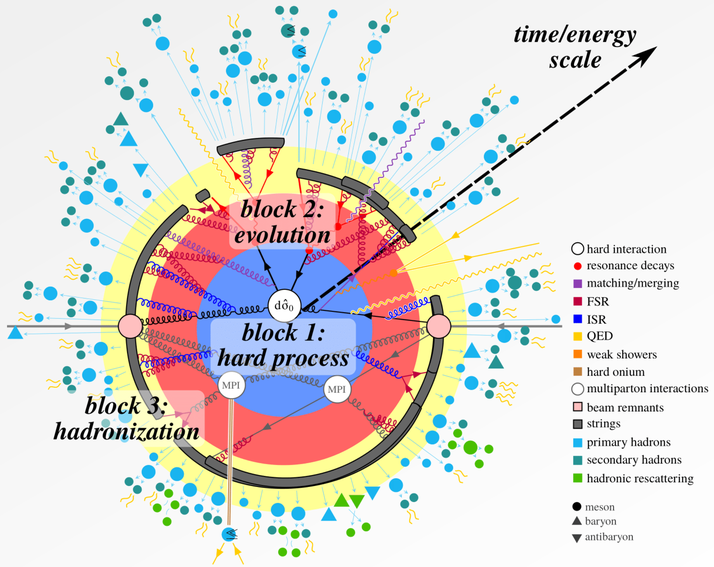
1.Motivation

Event generators simulate particle collisions in 3 blocks:

1) *Hard Process* - the initial high energy interaction (particle collsision) is calculated perturbatively.

2) *Parton Shower (Evolution)* – creation of quarks and gluons; it perturbatively evolves to the non-perturbative α_s scale.

3) *Hadronization* - quarks and gluons are combined into hadrons via phenomenological models.



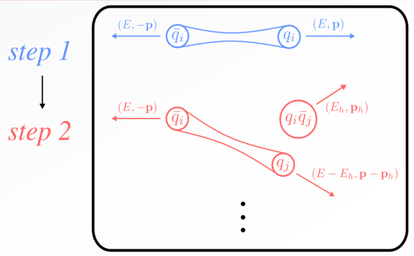
- The first two blocks can be calculated in perturbation theory, but hadronization is inherently non-perturbative and **cannot be calculated from first principals.**
- The state-of-the-art is to use phenomenological models with tuned parameters, which poorly describe the underlying physics.
- We propose to replace the parametrized model with our data-driven, Machine Learning based model, MLHad.

2. Lund String-Model

Currently, two models – the Cluster Model (CM) and Lund String Model (LSM) – are available to describe hadronization. The latter is used in Pythia and we consider a simplified version:

- A QCD color string connects the initial quark-antiquark $q_i\bar{q}_i$ pair, with energy E, carrying the 3 momentum $\pm p$ and iteratively splits to a hadron composed of $q_i\bar{q}_j$.
- Selection of flavor (FS) and Hadron kinematics are independent until the final stages.

Our model MLHad, uses machine learning to model each step.

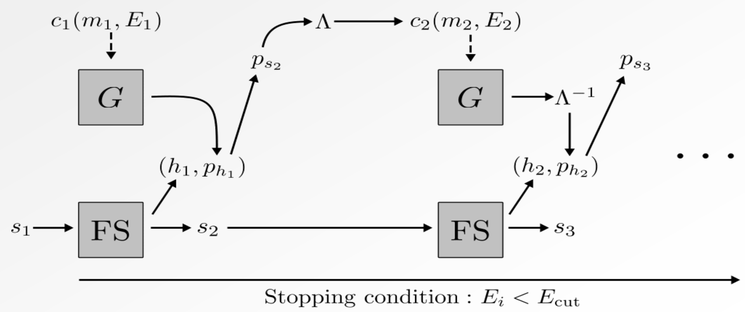


3. MLHad-Architecture

1) FS generates the flavor compositon of the hadron h_1 and new string end s_2.

2) Generator (G) generates the hadron kinematics p_{h_1}.

3) The produced hadron is stored and the string is updated. Step 1 is repeated.



4. Bayesian Normalizing Flows (BNFs)

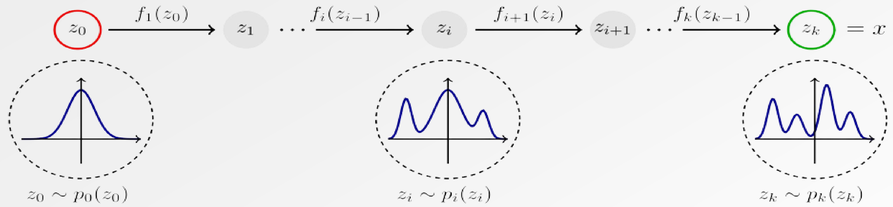


Image taken from https://github.com/janosh/awesome-normalizing-flows

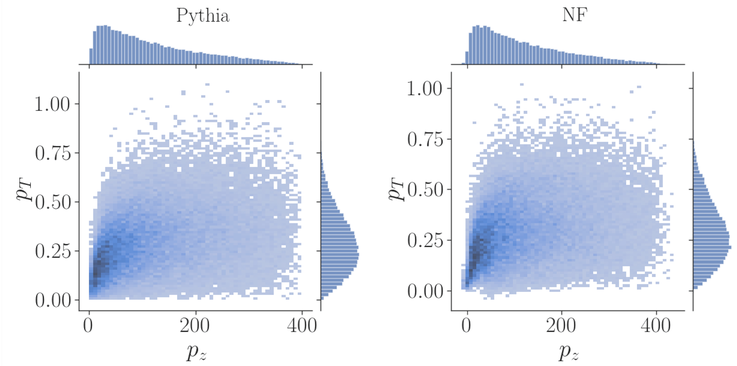
- NFs perform a bijective transformation by sampling a random vector z_0 from a simple (usually gaussian) distribution $p_0(z_0)$.
- The sample is transformed by a bijective mapping $z_0 = f^{-1}(z_k)$ to obtain a data point x from the desired (usually more complicated) distribution:

$$p_k(z_k) = p_0(z_0)\left|\det\left(\frac{\partial f}{\partial z_0}\right)\right|^{-1}$$

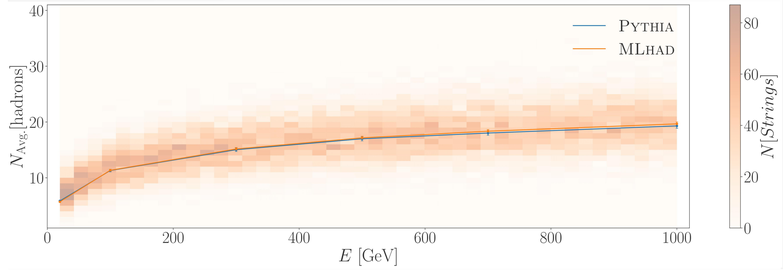
- For uncertainty estimation, we train a bayesian NF on the kinematics p_z and p_T of the emitted hadron.
- BNF is conditioned on the initial string Energy and hadron mass $c_i(m, E)$.

5.Results

Correlation plot between p_z and p_T, for Pythia and MLHad (right)



Average number of hadrons produced in a fragmentation chain of as a function of the initial parton energy, MLHad vs Pythia.



6. Conclusion and Outlook

We presented the MLHad pipeline first in:

- "Modeling hadronization using machine learning," [arXiv:2203.04983]; public code: https://gitlab.com/uchep/mlhad
- MLHad reproduces key features of a simplified Pythia hadronization model.
- It learns correlations between p_z and p_T , while allowing different hadron emission.
- Future work include uncertainty estimation, absent in common hadronzation models, and training on experimental data.

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